Semantic Exploration Analysis

Analysis of Semantic Exploration Bins

PENPAL Project

2025-11-27

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1 1. Data Loading and Preprocessing

1.1 1.1 Load Data

1.2 1.2 Descriptive Statistics

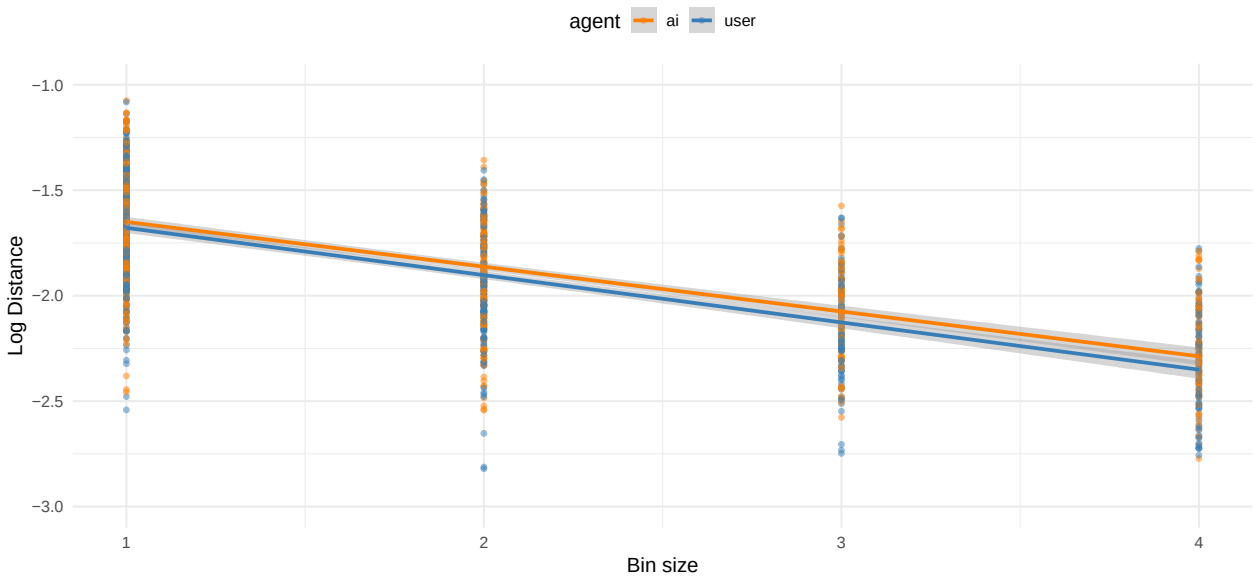
Table 1: Descriptive Statistics for Semantic Exploration by Agent

agent	n	mean_distance	sd_distance	mean_log_distance	sd_log_distance
ai	656	0.168	0.054	-1.839	0.337
interleaved	1886	0.140	0.053	-2.040	0.401
user	656	0.160	0.055	-2.104	2.745

2 2. Semantic Exploration Analysis

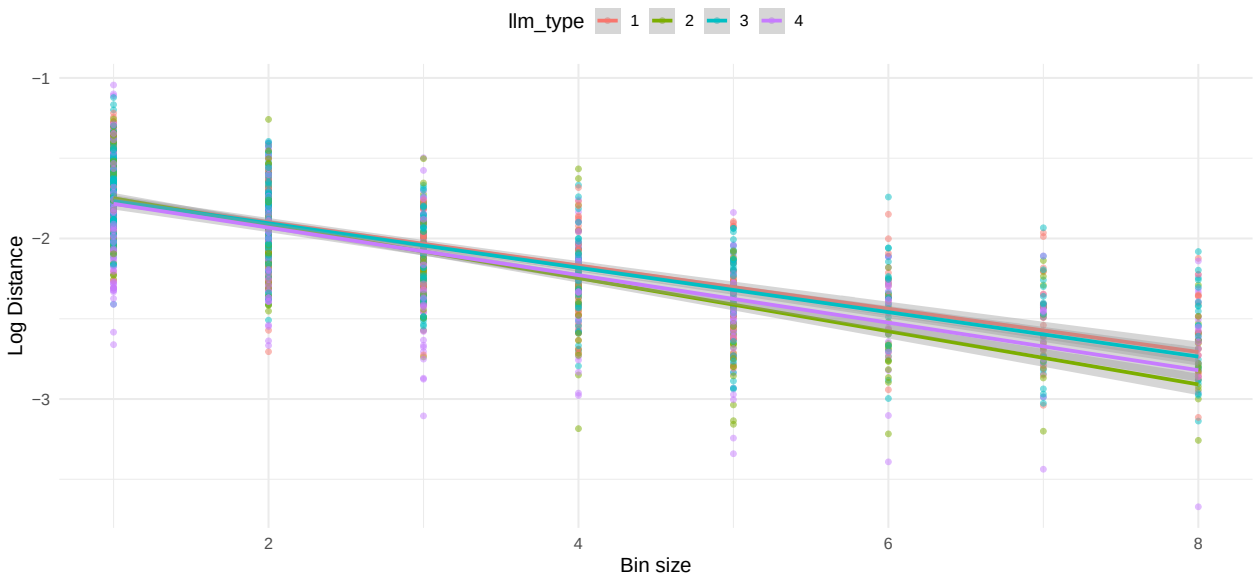
2.1 2.1 Bin Size vs. Distance Plot

Bin Size vs. Distance

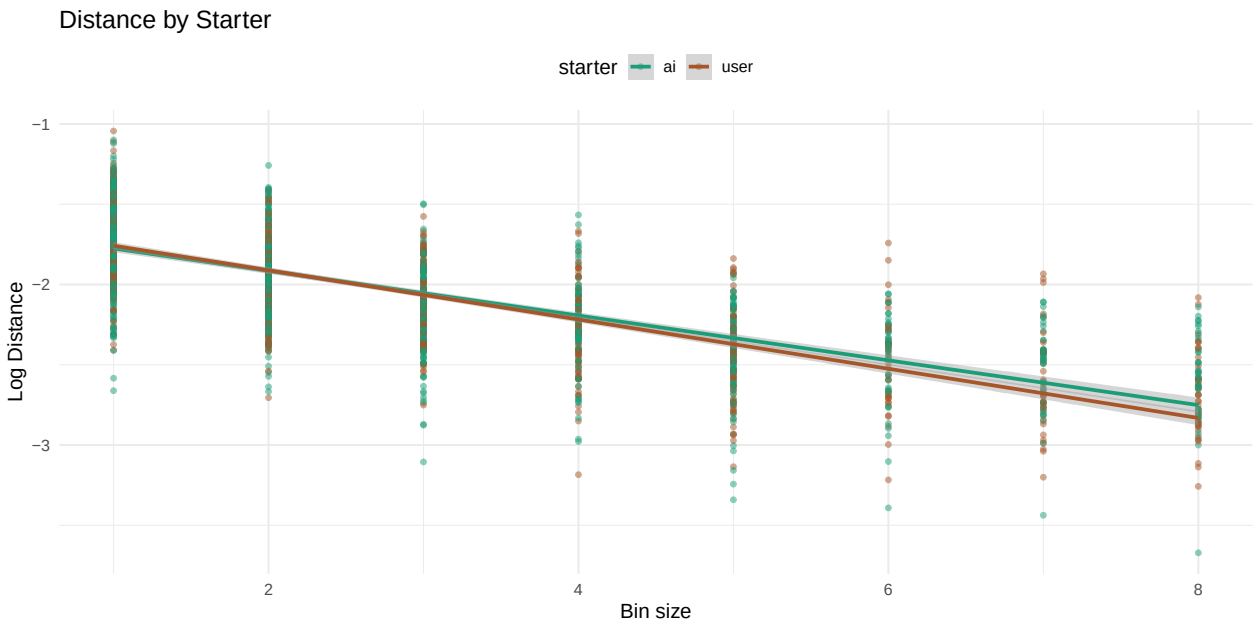


2.2 2.2 Interleaved Distance Plot

Distance (Interleaved)



2.3 2.3 Starter Effect Plot



3 3. Statistical Models

3.1 3.1 Agent Interaction Model

```
## === Agent Interaction Model ===

## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
## lmerModLmerTest]
## Formula: distance_log ~ k * agent + (1 | conversation_id)
##    Data: lm_agent
##
## REML criterion at convergence: 5270
##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
## -15.8708  -0.0862   0.0085   0.1183   4.6681
##
## Random effects:
##  Groups          Name        Variance Std.Dev.
##  conversation_id (Intercept) 0.923    0.9607
##  Residual                    2.883    1.6981
## Number of obs: 1312, groups:  conversation_id, 82
##
## Fixed effects:
##               Estimate Std. Error         df t value Pr(>|t|)
## (Intercept)   -1.44507    0.17199  357.65492  -8.402 1.05e-15 ***
## k             -0.21012    0.06295 1227.00000  -3.338 0.000869 ***
## agentuser     -0.44264    0.19144 1227.00000  -2.312 0.020934 *
## k:agentuser    0.09468    0.08902 1227.00000   1.064 0.287706
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
```

```
## Correlation of Fixed Effects:
##             (Intr) k      agntsr
## k           -0.686
## agentuser   -0.557  0.616
## k:agentuser  0.485 -0.707 -0.872
```

3.2 3.2 LLM Type Interaction Model (Interleaved)

```
## === LLM Type Interaction Model (Interleaved) ===

## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
## lmerModLmerTest]
## Formula: distance_log ~ k * factor(llm_type) + (1 | conversation_id)
##    Data: lm_llm_type
##
## REML criterion at convergence: -139.7
##
## Scaled residuals:
##     Min      1Q  Median      3Q     Max
## -3.1660 -0.6815  0.0355  0.6787  3.8356
##
## Random effects:
##  Groups          Name        Variance Std.Dev.
##  conversation_id (Intercept) 0.02415  0.1554
##  Residual                    0.04750  0.2179
## Number of obs: 1886, groups:  conversation_id, 82
##
## Fixed effects:
##                       Estimate Std. Error         df t value Pr(>|t|)
## (Intercept)         -1.629e+00  4.100e-02  1.038e+02 -39.726  < 2e-16 ***
## k                   -1.348e-01  5.226e-03  1.800e+03 -25.795  < 2e-16 ***
## factor(llm_type)2    4.466e-02  5.799e-02  1.038e+02   0.770   0.4430
## factor(llm_type)3    3.558e-04  5.424e-02  1.038e+02   0.007   0.9948
## factor(llm_type)4   -7.103e-03  5.529e-02  1.038e+02  -0.128   0.8980
## k:factor(llm_type)2 -3.079e-02  7.391e-03  1.800e+03  -4.166 3.25e-05 ***
## k:factor(llm_type)3 -3.606e-03  6.914e-03  1.800e+03  -0.522   0.6020
## k:factor(llm_type)4 -1.325e-02  7.047e-03  1.800e+03  -1.880   0.0602 .
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##             (Intr) k      fc(_)2 fc(_)3 fc(_)4 k:(_)2 k:(_)3
## k           -0.366
## fctr(llm_)2 -0.707  0.259
## fctr(llm_)3 -0.756  0.276  0.535
## fctr(llm_)4 -0.742  0.271  0.524  0.561
## k:fctr(l_)2  0.259 -0.707 -0.366 -0.196 -0.192
## k:fctr(l_)3  0.276 -0.756 -0.196 -0.366 -0.205  0.535
## k:fctr(l_)4  0.271 -0.742 -0.192 -0.205 -0.366  0.524  0.561
```

3.3 3.3 Starter Interaction Model (Interleaved)

```
## === Starter Interaction Model (Interleaved) ===

## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
```

```
## lmerModLmerTest]
## Formula: distance_log ~ k * factor(starter) + (1 | conversation_id)
##    Data: lm_starter
##
## REML criterion at convergence: -149.5
##
## Scaled residuals:
##     Min      1Q  Median      3Q     Max
## -3.2197 -0.6612  0.0307  0.6784  3.6096
##
## Random effects:
##  Groups          Name        Variance Std.Dev.
##  conversation_id (Intercept) 0.02386  0.1545
##  Residual                    0.04780  0.2186
## Number of obs: 1886, groups:  conversation_id, 82
##
## Fixed effects:
##                         Estimate Std. Error         df t value Pr(>|t|)
## (Intercept)           -1.636e+00  2.673e-02  1.070e+02 -61.208  < 2e-16 ***
## k                     -1.394e-01  3.432e-03  1.802e+03 -40.608  < 2e-16 ***
## factor(starter)user    3.134e-02  3.827e-02  1.070e+02   0.819  0.41469
## k:factor(starter)user -1.393e-02  4.914e-03  1.802e+03  -2.835  0.00463 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##             (Intr) k      fctr()
## k           -0.368
## fctr(strtr) -0.698  0.257
## k:fctr(str)  0.257 -0.698 -0.368
```

4 4. Session Information

```
## R version 4.4.2 (2024-10-31)
## Platform: aarch64-apple-darwin20
## Running under: macOS Sequoia 15.6.1
##
## Matrix products: default
## BLAS:   /Library/Frameworks/R.framework/Versions/4.4-arm64/Resources/lib/libRblas.0.dylib
## LAPACK: /Library/Frameworks/R.framework/Versions/4.4-arm64/Resources/lib/libRlapack.dylib;  LAPACK ve
##
## locale:
## [1] en_US.UTF-8/en_US.UTF-8/en_US.UTF-8/C/en_US.UTF-8/en_US.UTF-8
##
## time zone: Europe/Copenhagen
## tzcode source: internal
##
## attached base packages:
## [1] stats     graphics  grDevices utils     datasets  methods   base
##
## other attached packages:
##  [1] emmeans_1.10.6  here_1.0.1      afex_1.4-1      cowplot_1.1.3
##  [5] patchwork_1.3.0 zoo_1.8-12      stringi_1.8.4   arrow_19.0.1.1
```

```
##  [9] lmerTest_3.1-3  lme4_1.1-35.5   Matrix_1.7-1    lubridate_1.9.3
## [13] forcats_1.0.0   stringr_1.5.1   dplyr_1.1.4     purrr_1.0.2
## [17] readr_2.1.5     tidyr_1.3.1     tibble_3.2.1    ggplot2_3.5.1
## [21] tidyverse_2.0.0
##
## loaded via a namespace (and not attached):
##  [1] tidyselect_1.2.1    farver_2.1.2        fastmap_1.2.0
##  [4] TH.data_1.1-2       pacman_0.5.1        digest_0.6.37
##  [7] timechange_0.3.0    estimability_1.5.1  lifecycle_1.0.4
## [10] survival_3.7-0      magrittr_2.0.3      compiler_4.4.2
## [13] rlang_1.1.4         tools_4.4.2         utf8_1.2.4
## [16] yaml_2.3.10         knitr_1.50          labeling_0.4.3
## [19] bit_4.0.5           plyr_1.8.9          abind_1.4-8
## [22] multcomp_1.4-26     withr_3.0.2         numDeriv_2016.8-1.1
## [25] grid_4.4.2          fansi_1.0.6         xtable_1.8-4
## [28] colorspace_2.1-1    scales_1.3.0        MASS_7.3-61
## [31] tinytex_0.52        cli_3.6.3           mvtnorm_1.3-3
## [34] rmarkdown_2.28      ragg_1.3.2          generics_0.1.3
## [37] rstudioapi_0.16.0   reshape2_1.4.4      tzdb_0.4.0
## [40] minqa_1.2.8         splines_4.4.2       assertthat_0.2.1
## [43] parallel_4.4.2      vctrs_0.6.5         boot_1.3-31
## [46] sandwich_3.1-1      carData_3.0-5       car_3.1-3
## [49] hms_1.1.3           bit64_4.0.5         Formula_1.2-5
## [52] systemfonts_1.1.0   glue_1.7.0          nloptr_2.1.1
## [55] codetools_0.2-20    gtable_0.3.5        munsell_0.5.1
## [58] pillar_1.9.0        htmltools_0.5.8.1   R6_2.5.1
## [61] textshaping_0.4.0   rprojroot_2.0.4     evaluate_1.0.3
## [64] lattice_0.22-6      Rcpp_1.0.13         coda_0.19-4.1
## [67] nlme_3.1-166        mgcv_1.9-1          xfun_0.54
## [70] pkgconfig_2.0.3
```

Analysis complete. All figures saved to `analysis/figures/` and tables to `analysis/tables/`.